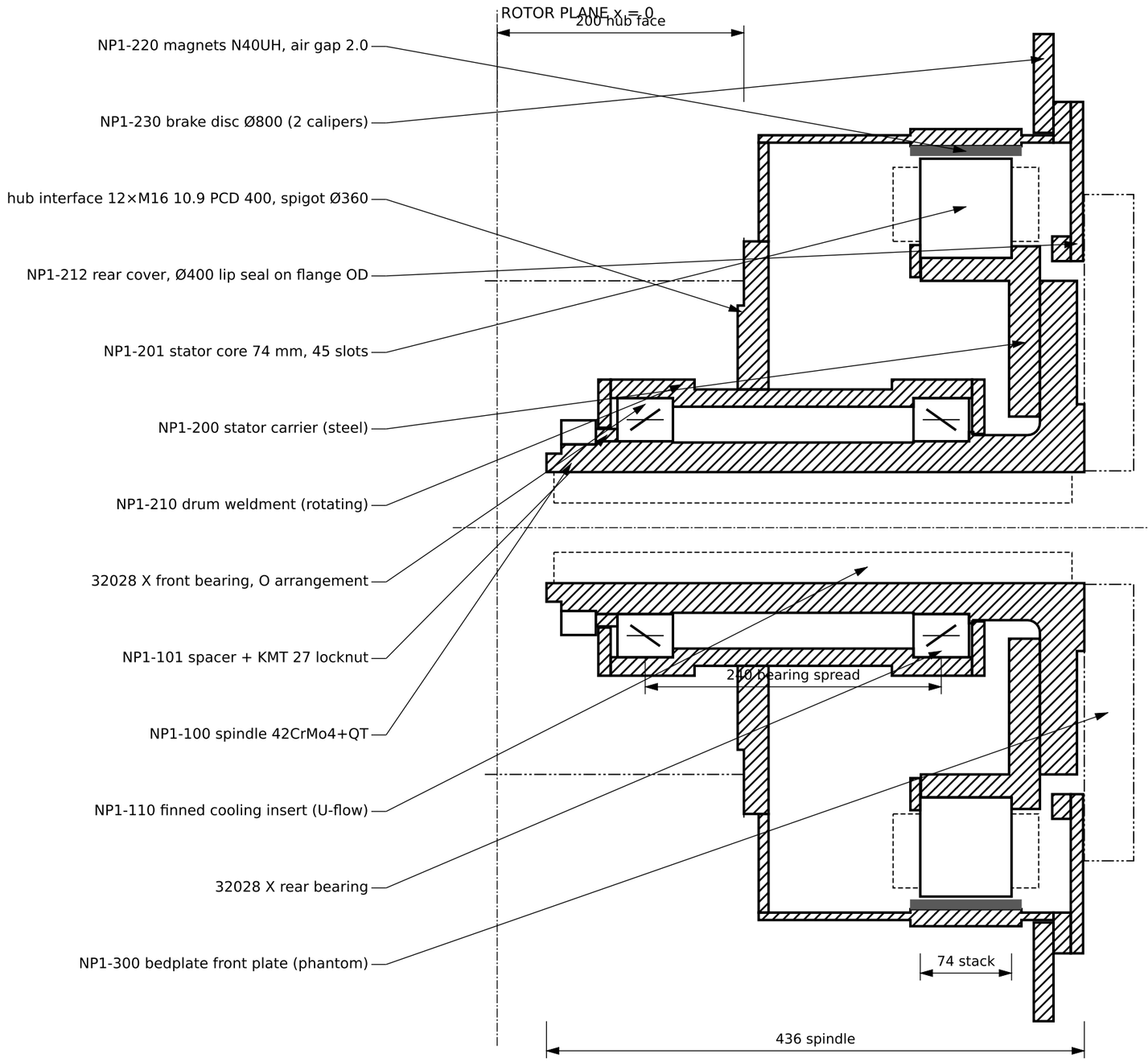
SECTION ON SPINDLE AXIS (both halves sectioned)



Item	Part	Qty	Material / type
1	NP1-100 spindle	1	42CrMo4+QT forged
2	NP1-101 spacer	1	42CrMo4, induction hard.
3	32028 X bearing	2	bought
4	KMT 27 locknut	1	bought
5	NP1-110 cooling insert	1	6082-T6 extrusion
6	NP1-200 stator carrier	1	S355J2+N
7	NP1-201 stator core	≈205 sheets	M270-35A bonded
8	NP1-202 clamp ring	1	1.4404
9	NP1-210 drum weldment	1	S355J2+N
10	NP1-211 seal covers	2	S355J2
11	NP1-212 rear cover	1	5083-H111
12	NP1-220 magnets	160	N40UH bought
13	NP1-230 brake disc	1	C45E+N

NOTES

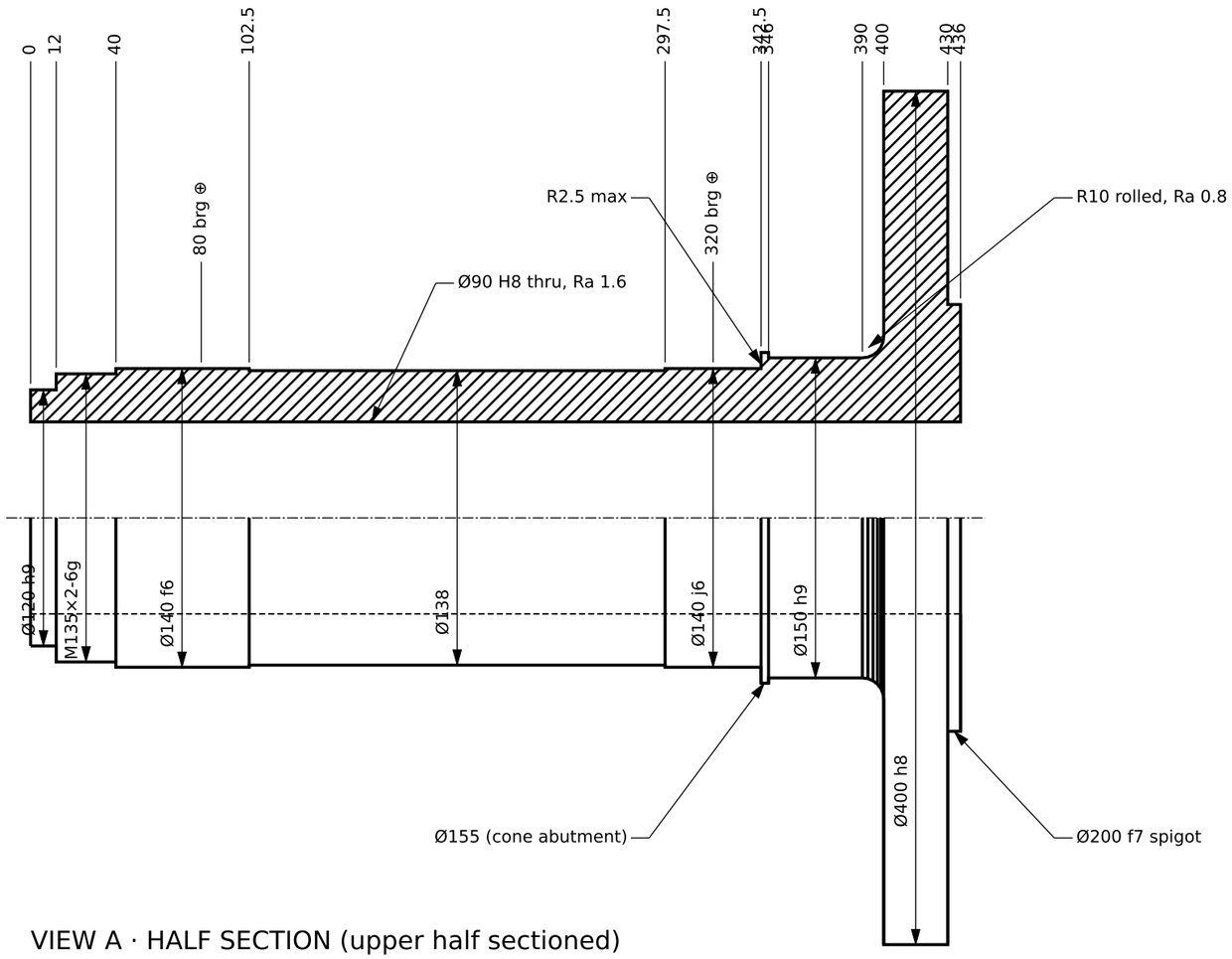
1. x is measured rearward from the rotor plane along the spindle axis; the spindle axis is tilted 5° nose up (bedplate NP1-300).
2. Stationary: spindle, cones, stator carrier, stator, cooling insert, bedplate. Rotating: drum, cups, seal covers, rear cover, magnets, brake disc, hub.
3. Bearings: back-to-back (O) tapered roller pair, preload 2 to 4 kN (end play −0.02 to −0.05) set by the KMT 27 locknut, checked by rolling torque before seals are fitted (doc 14, 5.3).
4. Air gap 2.0 mm nominal; eccentricity ≤ 0.20 mm after assembly, measured with 4 capacitive gap sensors (doc 15).
5. Sealed generator volume: front by the hub, rear by the Ø400 lip seal + labyrinth, desiccant breather on the rear cover.
6. Hub, pitch system, nacelle cover and slip ring are not part of P1 bench article (dummy hub NP1-600 instead).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

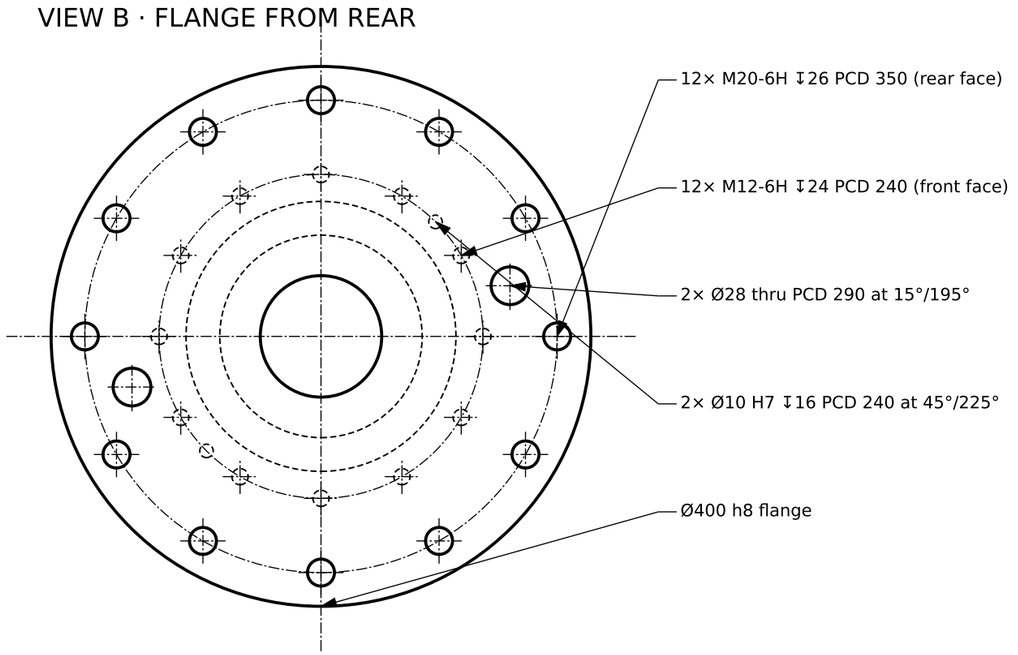
12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

GENERAL ARRANGEMENT · DRIVETRAIN SECTION

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-000	A · 2026-11-02	1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see parts list	≈ 320 kg (drivetrain, no hub)	Claude / not checked



VIEW A · HALF SECTION (upper half sectioned)



VIEW B · FLANGE FROM REAR

NOTES

1. Blank: open-die forging 42CrMo4, forging ratio ≥ 3, quenched and tempered to Rp0.2 ≥ 650 MPa, Rm 900 to 1100 MPa; KV ≥ 27 J at −40 °C (3 specimens, longitudinal, taken from the flange zone). Certificate EN 10204 3.1.
2. Ultrasonic test of the blank to EN 10228-3 quality class 3; MT of the finished part (fillets, shoulders, thread) to EN ISO 9934, no linear indications.
3. Bearing seats: Ø140 f6 front (cone must slide for preload), Ø140 j6 rear; Ra 0.8; cylindricity 0.006; run-out of both seats and of Ø400 and Ø200 to common axis A-B ≤ 0.01.
4. Flange rear face and front face perpendicular to A-B within 0.02; flatness 0.02. Ø400 seal surface Ra 0.2 to 0.4, plunge ground (no lead), hard chrome 30 µm or wear sleeve.
5. Thread M135×2-6g with undercut DIN 76-D; precision locknut KMT 27 (or equivalent with locking screws).
6. Bore Ø90 H8 for the cooling insert NP1-110 (installed with liquid nitrogen shrink, interference 0.02 to 0.06).
7. Two radial holes Ø6 from the bore to under each bearing cone seat at 90° (bottom dead centre at 0°) for PT100 bearing temperature sensors; deburr, edges R0.5.
8. All fillets and the shoulders rolled or shot peened; no stamping on the part: mark serial number on the flange OD rear chamfer by etching.
9. Protect: phosphate + oil in storage; seats and seal surfaces taped until assembly.

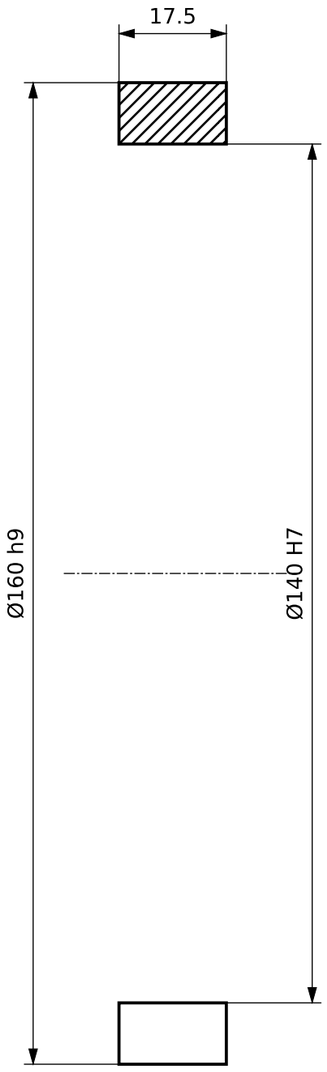
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

SPINDLE (STATIONARY MAIN SHAFT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-100	A · 2026-11-02	1:2.86 (A) / 1:4.5 (B)
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
42CrMo4+QT (EN 10083-3), forged	56.7 kg	Claude / not checked

SECTION



NOTES

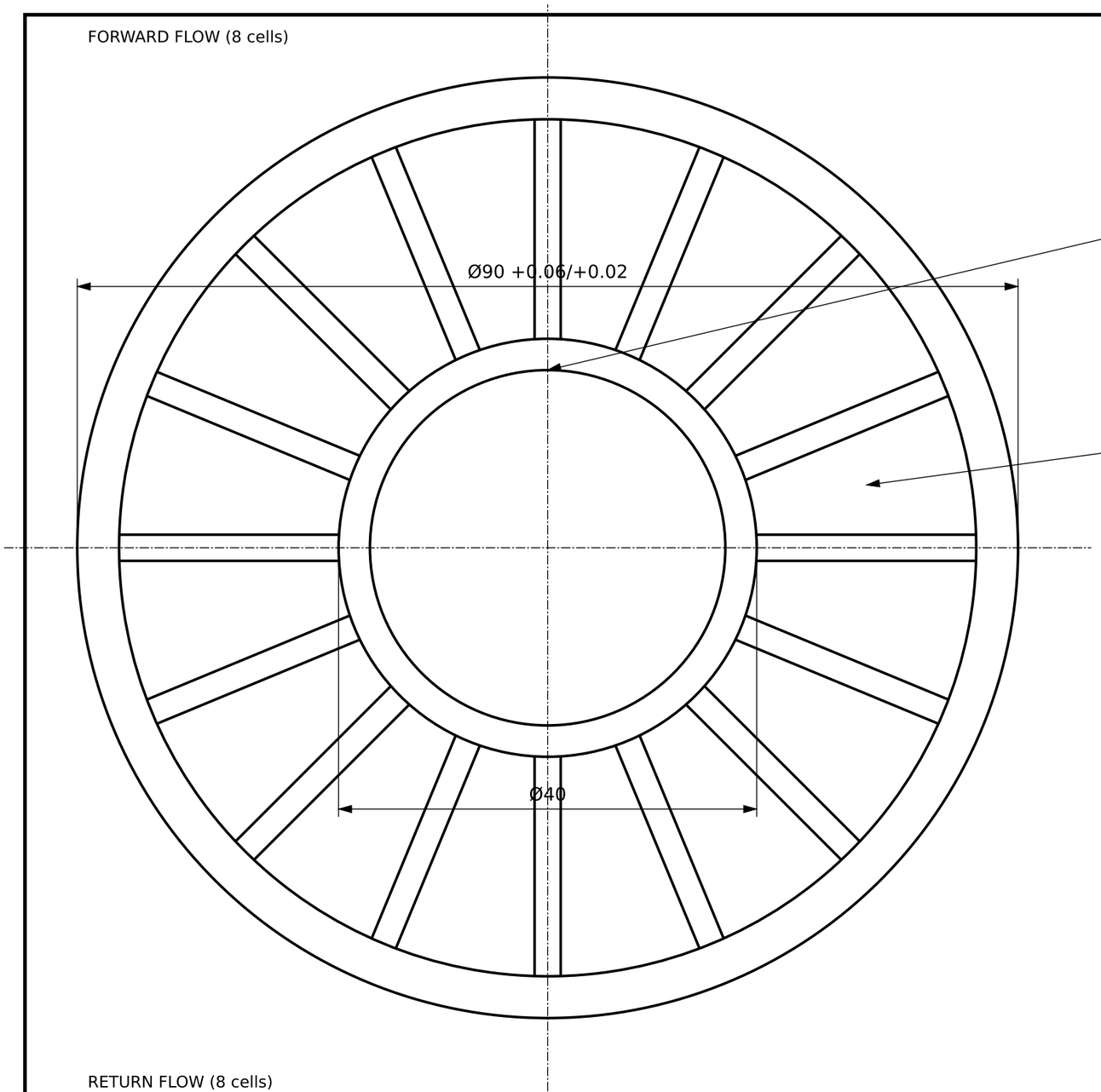
1. Faces parallel within 0.005 (it sets the bearing preload path).
2. Ø160 surface induction hardened 55 to 60 HRC, depth 1 mm, ground Ra 0.2 to 0.4, no lead.
3. Bore chamfer 1.5×30° on the cone side.

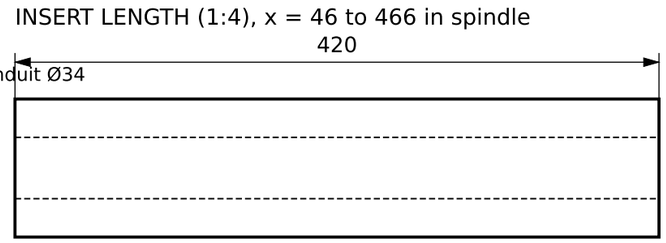
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

FRONT SPACER (LIP SEAL COUNTERFACE)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-101	A · 2026-11-02	1:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
42CrMo4+QT, surface induction hardened	0.6 kg	Claude / not checked





NOTES

1. The two fins at 0° and 180° are the flow partition: they are continuous and sealed to the front plenum cap so that air flows forward in the upper 8 cells and returns in the lower 8 (U-flow, both ports at the rear).
2. Front plenum cap (6082, 10 mm, O-ring to bore) closes the cells 10 mm from the insert front end and lets the Ø34 conduit pass through to the slip ring position.
3. Shrink into spindle bore with liquid nitrogen; interference 0.02 to 0.06 at 20 °C gives contact pressure in operation (Al expands more than steel). Thermal paste is not used (shrink fit).
4. Profile tolerances EN 755-9; cut length ±0.5; deburr; no anodising on the OD (thermal contact).
5. Rear manifold (fan, filter, inlet/outlet split) bolts to the spindle rear face, see NP1-300 and doc 14 section 6.

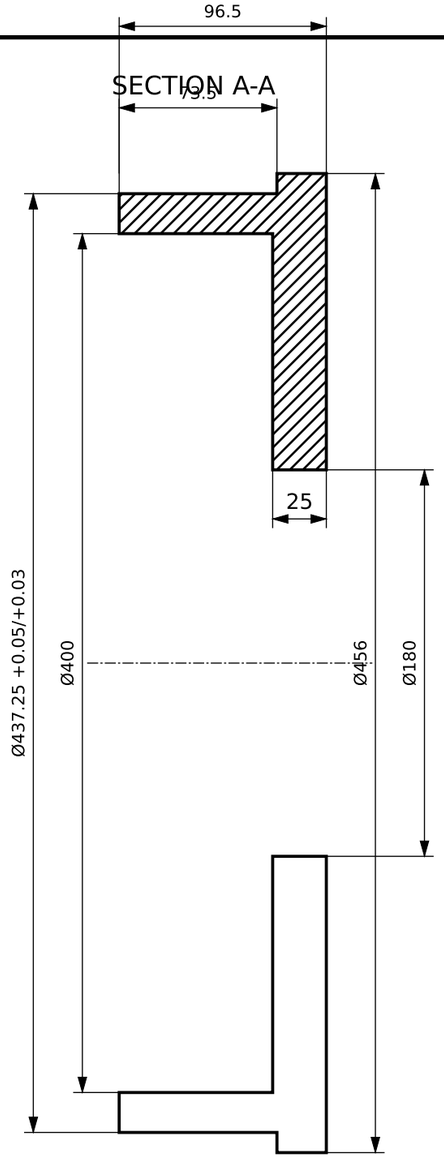
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

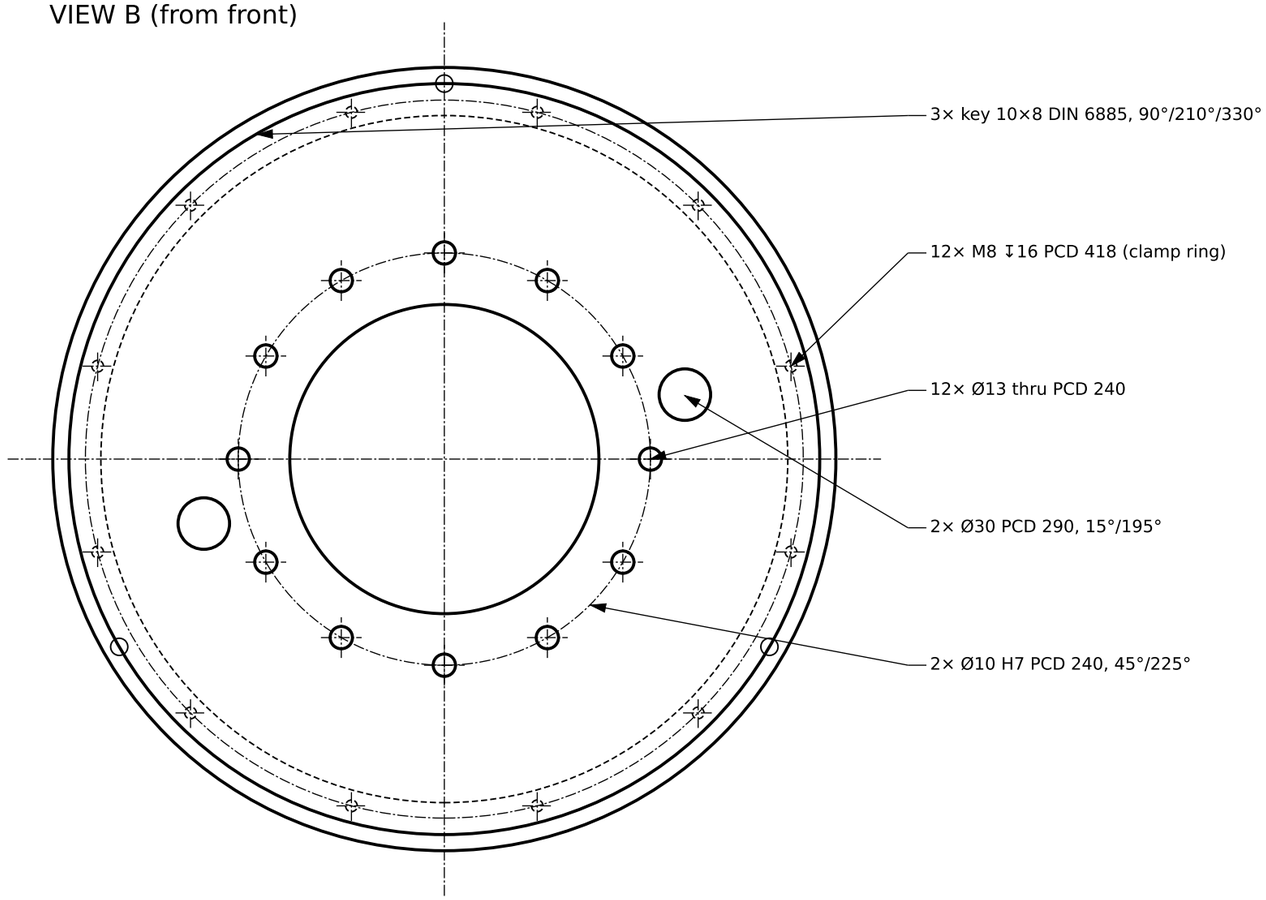
SPINDLE COOLING INSERT (FINNED EXTRUSION)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-110	A · 2026-11-02	2:1 / 1:4
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
EN AW-6082-T6 extrusion	2.6 kg	Claude / not checked

SECTION A-A



VIEW B (from front)



NOTES

1. Steel, not aluminium, is used for P1: steel matches the lamination expansion, so the stack stays tight and centred from −40 °C to +130 °C. An aluminium carrier loses ≈ 0.6 mm of fit on Ø437 at −40 °C (doc 14, 4.2).
2. Stack seat Ø437.25 +0.05/+0.03 against lamination Ø437.20 +0.03/0: interference 0 to 0.05; assemble by cooling the carrier (dry ice / LN2) and sliding the bonded stack on against the rear stop.
3. Run-out of the stack seat and of the rear stop face to the bolting face and Ø180 spigot ≤ 0.02. Bolting face flatness 0.02.
4. Rim front face sits 0.5 below the stack face so the clamp ring NP1-202 clamps the stack (12× M8 A4-80, 20 Nm).
5. Bolt to the spindle flange with 12× M12 10.9 + hardened washers, 110 Nm (μ 0.12), with 2 fitted dowels; thermal interface: Al-filled paste, 0.05 mm, on the full bolting face.
6. Machine after stress relief (580 to 620 °C, 2 h). Paint: none on the stack seat and bolting face; elsewhere epoxy primer 60 μm.

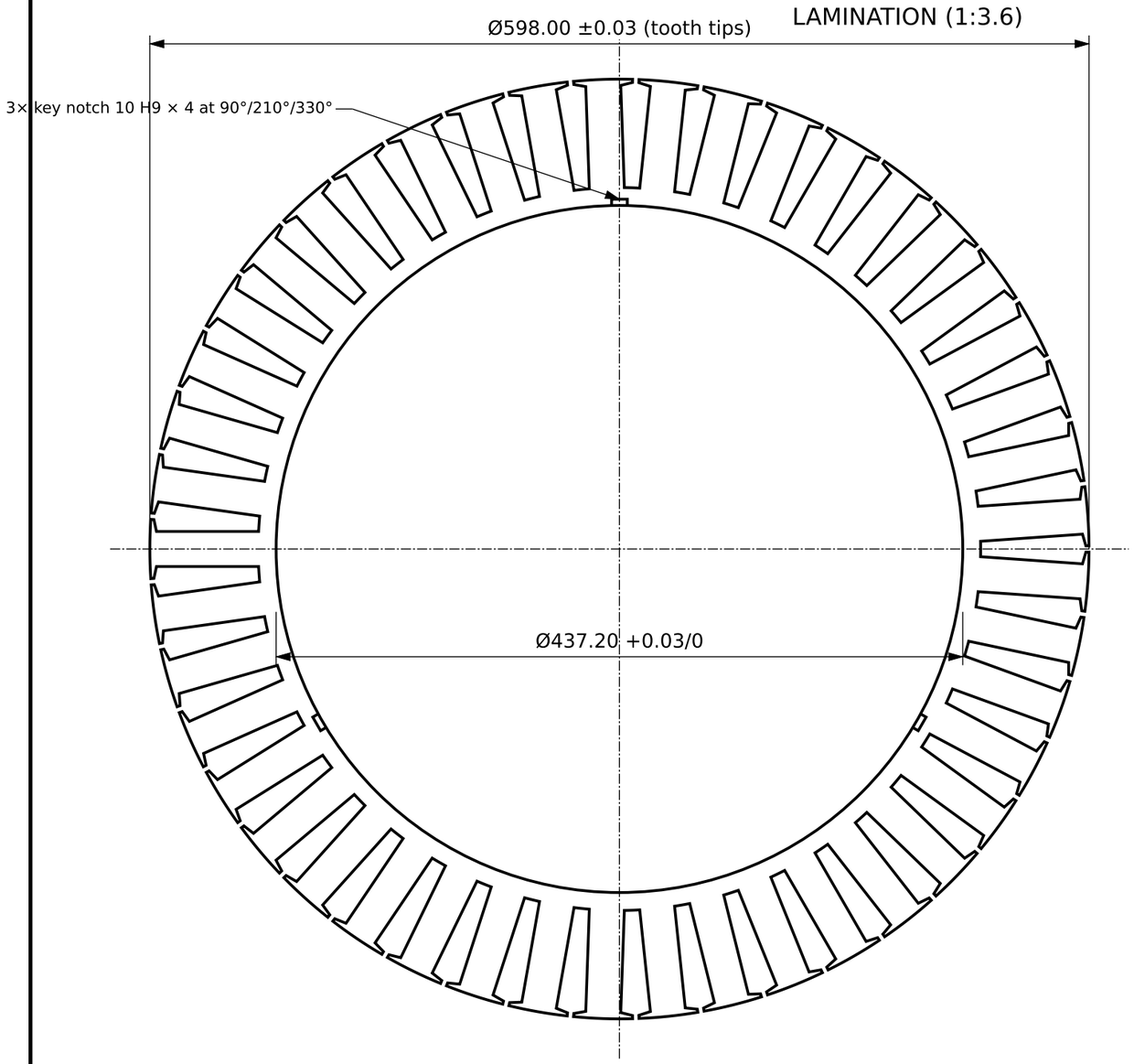
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

STATOR CARRIER

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-200	A · 2026-11-02	1:2.86 / 1:3.6
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N plate or ring forging, stress relieved	40.6 kg	Claude / not checked

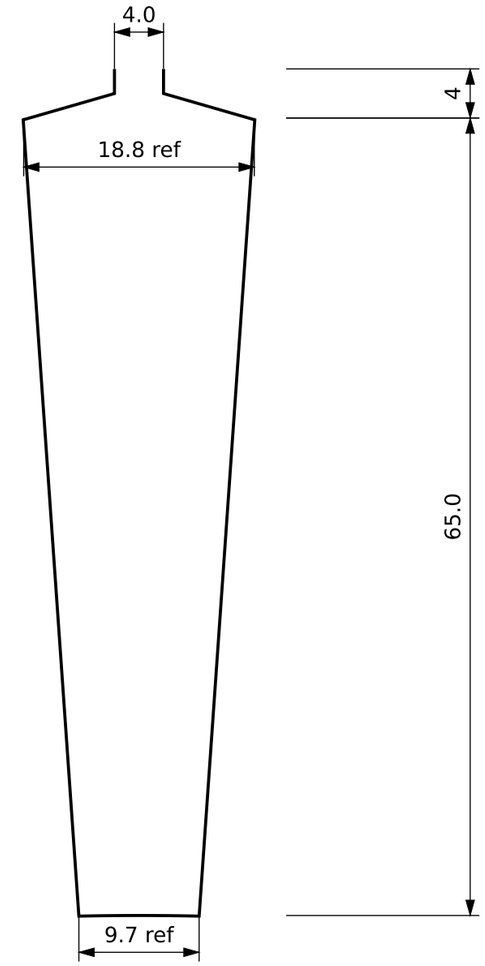
LAMINATION (1:3.6)



SLOT DETAIL (2:1)

tooth body 22.3 parallel-sided

slot area 932 mm² (FEA basis)



NOTES

1. DXF of this sheet (layer OUTLINE) is the cutting contour. Production: stamping die or laser + edge-anneal; P1: laser cut, burr ≤ 0.02 mm, cut-edge heat zone ≤ 0.1 mm.
2. Material M270-35A (EN 10106) with self-bonding varnish (Backlack, e.g. class C coating, both sides). Grain orientation irrelevant (non-oriented), but rotate each sheet 3 slot pitches (24°) when stacking to average thickness variation.
3. Stack: ≈ 205 sheets for 74.0 mm at stacking factor ≥ 0.97; bond-cure in fixture on a Ø437.20 mandrel with 3 keys, 1.0 to 1.5 MPa axial pressure.
4. Stack checks: length 74.0 ±0.2; Ø598.0 OD run-out ≤ 0.05 to the bore; slot alignment: Ø3.9 gauge pin passes every slot opening full length.
5. Interlaminar resistance: Franklin test to IEC 60404-11, ≥ 10 Ω·cm² after bonding.
6. Tooth tips are not machined after stacking (damages insulation). If OD run-out exceeds 0.05, reject the stack.

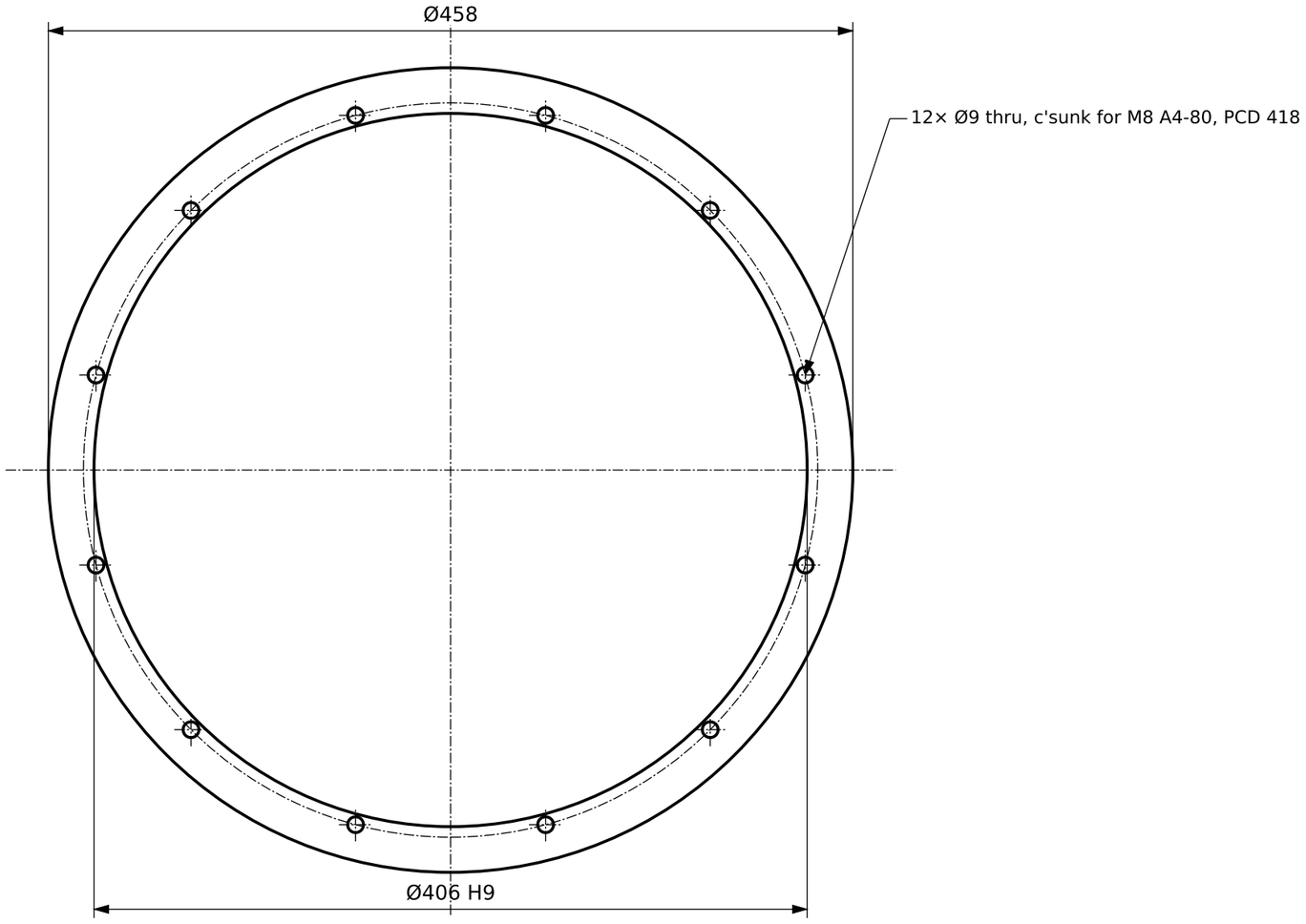
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

STATOR LAMINATION (45 SLOTS) · LASER / STAMPING

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-201	A · 2026-11-02	1:3.6 / 2:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
M270-35A, 0.35 mm, self-bonding varnish	48.0 kg per stack	Claude / not checked

VIEW (thickness 8.0, faces parallel 0.05)



NOTES

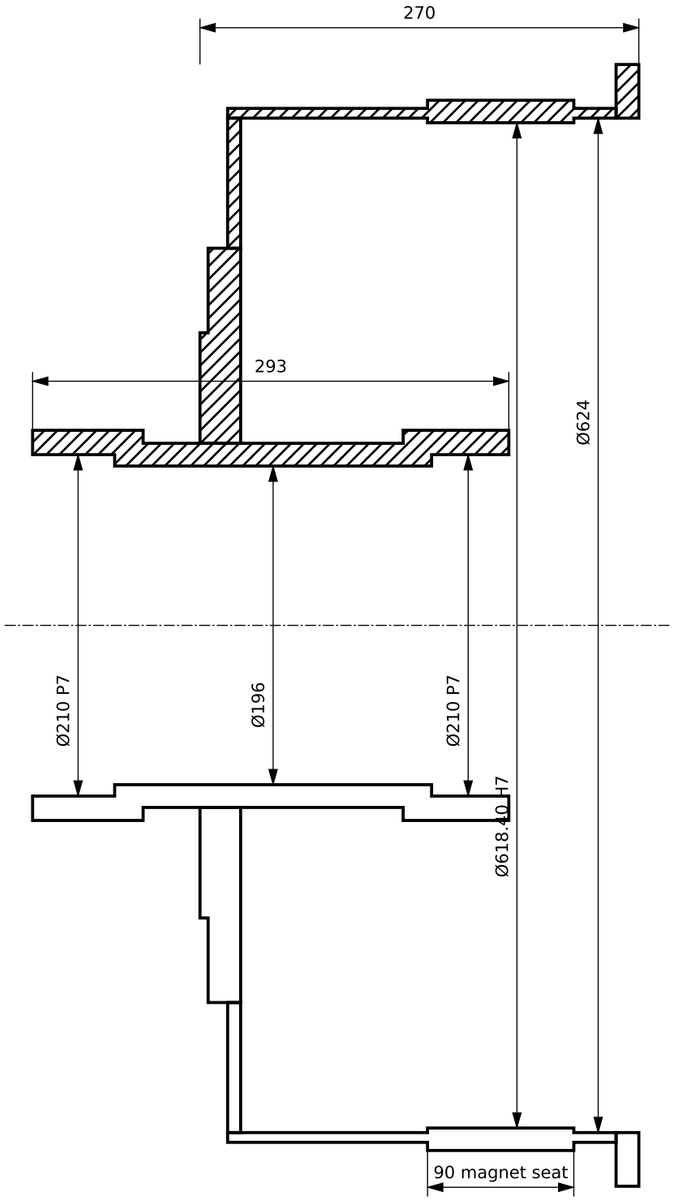
1. Laser cut from 8 mm plate, faces machined flat 0.05.
2. Must not touch the coils: inner end-winding radius ≥ 232; ring OD 229.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

STATOR FRONT CLAMP RING

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-202	A · 2026-11-02	1:3.3
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
1.4404 (non-magnetic)	2.2 kg	Claude / not checked



SECTION A-A (weld preps not shown, note 2)

Feature	Size	Tolerance / note
cup seats (2)	Ø210 × 45	P7, Ra 0.8, coax. 0.02
cup shoulders	Ø196	square 0.01 to axis
housing OD	Ø240 / Ø224	seal cover faces at x 92, 385
hub spigot	Ø360 × 5	f7; face flat 0.03
hub web	20 thick, Ø464	12× M16 thru PCD 400
outer web	8 thick	full pen. weld to shell
magnet seat	Ø618.40 × 90	H7, coax. 0.02 to cups
shell OD	Ø646 / Ø636	magnet zone 13.8 wall
shell ID (free)	Ø624	clearance to magnets
rear flange ring	Ø624/Ø690 × 14	M10 PCD 644, M12 PCD 662
lock holes	12× Ø26 H8 ↧20	PCD 662, interleaved

NOTES

1. Parts: housing tube (seamless S355J2H or ring forging), hub web 20 mm, outer web 8 mm, shell (rolled 16 mm plate, one longitudinal full-penetration weld, or ring rolled), rear flange ring 14 mm.
2. Welds: full penetration where the web meets the housing and the shell; ISO 5817 class B; 100 % MT, UT on the shell seam. Stress relieve 580 to 620 °C before machining.
3. Machine in one set-up after stress relief: cup seats Ø210 P7, magnet seat Ø618.40 H7, hub spigot Ø360 f7, rear flange face. Coaxiality of both cup seats and magnet seat ≤ 0.02; cup shoulder faces square to axis ≤ 0.01.
4. Hub web: 12× M16-6H through, PCD 400 (doc 13 joint, 10.9 preloaded). Rear flange ring: 24× M10-6H ↧20 PCD 644 (rear cover) and 24× M12-6H ↧20 PCD 662 + 12× Ø26 H8 ↧20 lock holes PCD 662 interleaved (36 positions at 10°).
5. Front and rear housing end faces: 6× M8 ↧16 on PCD 225 for the seal covers NP1-211; 2 grease nipples M10×1 into each bearing space with purge holes opposite.
6. Balance the finished rotor (drum, magnets, disc) to ISO 21940-11 G2.5 at 250 rpm, correction on the rear flange ring.
7. Magnet seat: degrease, grit blast Sa 2½ just before bonding; no paint inside. Outside: zinc spray + epoxy (C5) after machining.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

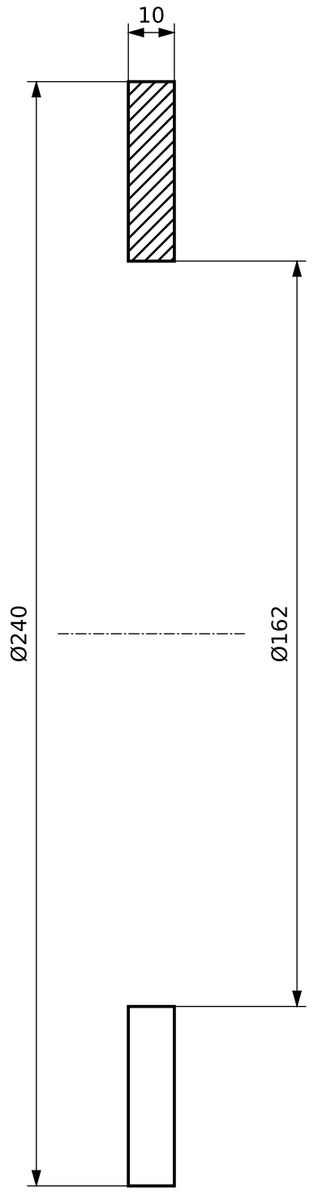
12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

ROTOR DRUM WELDMENT (GENERATOR YOKE + BEARING HOUSING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-210	A · 2026-11-02	1:3.8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N, welded, stress relieved	96.0 kg	Claude / not checked

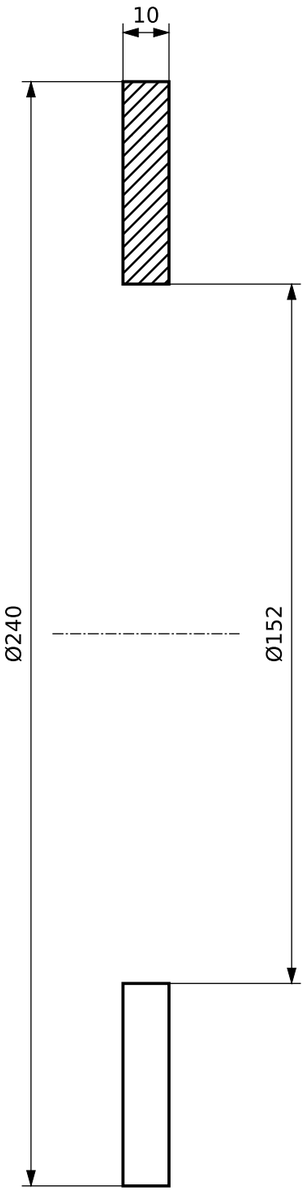
FRONT COVER: lip seal 160×190×10 HNBR + V-ring

6× Ø9 PCD 225; seal bore H8, Ra 1.6



REAR COVER: lip seal 150×180×10 HNBR

6× Ø9 PCD 225; seal bore H8, Ra 1.6



NOTES

1. Seal lips face the bearing (grease retention); V-ring outside on the front cover runs against the spacer face.
2. Purge: grease exits under the lip; collect with absorbent ring at service.
3. Pilot on the housing bore Ø210, 2 mm deep (not shown), for concentric seals.

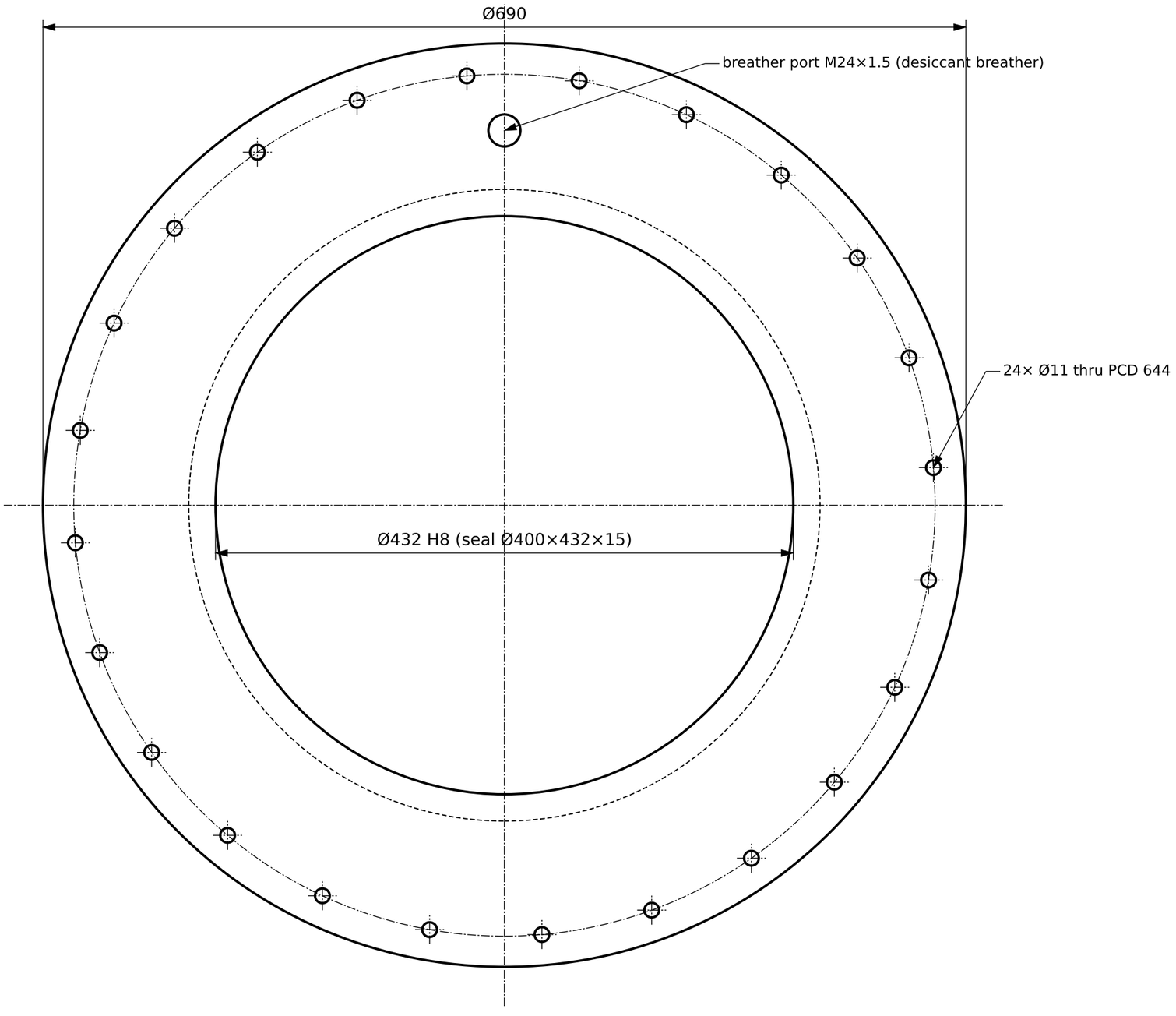
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

MAIN BEARING SEAL COVERS (FRONT / REAR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-211	A · 2026-11-02	1:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2	4.1 kg	Claude / not checked

VIEW FROM REAR · plate 10, seal ring 15 × 20 welded (x 450 to 465)



NOTES

1. Rear seal: radial lip seal Ø400×432×15 HNBR −40 °C (or split seal), plus labyrinth groove; runs on the spindle flange OD (3.7 m/s at 178 rpm).
2. Ring to plate: TIG fillet both sides, then machine seal bore concentric to the bolt-circle pilot within 0.05.
3. Inspection port Ø60 with cover at 180° (not shown) for air-gap feeler and sensor cables during P1 tests.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

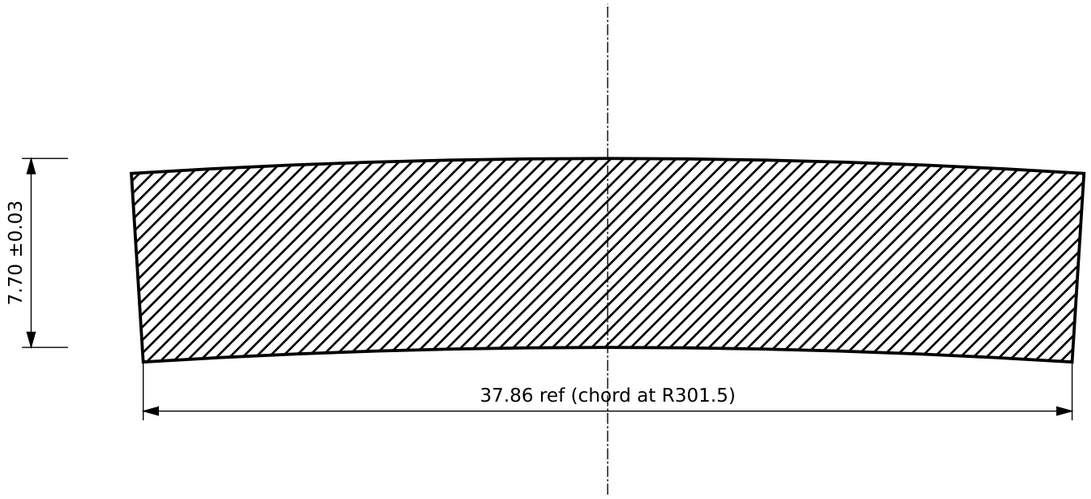
DRUM REAR COVER WITH SEAL CARRIER

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-212	A · 2026-11-02	1:3.6
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
EN AW-5083-H111 plate + ring	7.3 kg	Claude / not checked

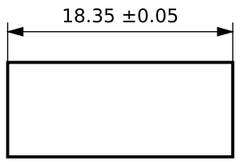
SECTION (4:1)

R309.20 −0.05/0 outer (bonded face), R301.50 inner, arc 7.20°

magnetised radially; pole marked on the outer face



SIDE (2:1)



NOTES

1. Grade N40UH (Br ≥ 1.25 T, HcJ ≥ 1990 kA/m at 20 °C); supplier to give the demagnetisation curve at 20, 80, 120 and 150 °C (doc 05: HcJ ≥ 896 kA/m at 120 °C required).
2. Length 18.35 ±0.05 (4 pieces per pole, axial segmentation for eddy loss; 0.2 mm bond gaps). 160 pieces + 8 spares, half N-in, half S-in, marked on the outer face.
3. Coating NiCuNi 10 to 20 µm + epoxy 15 to 25 µm; salt spray ≥ 240 h (ISO 9227) on witness samples; corners R0.3.
4. Flux check: each piece within ±1.5 % of the batch mean (Helmholtz coil); reject outliers. Record the serial batch for traceability.
5. Bonding into the drum: structural epoxy rated 150 °C, positioned by a GRP comb (pole pitch 9.000°, 40 poles), then 0.5 mm glass/epoxy wet layup over the inner face (vacuum bag) as retention and corrosion barrier.
6. Handling: magnets must never approach a steel surface uncontrolled; magnet insertion jig with brass slider and screw feed (doc 14, 7.2).

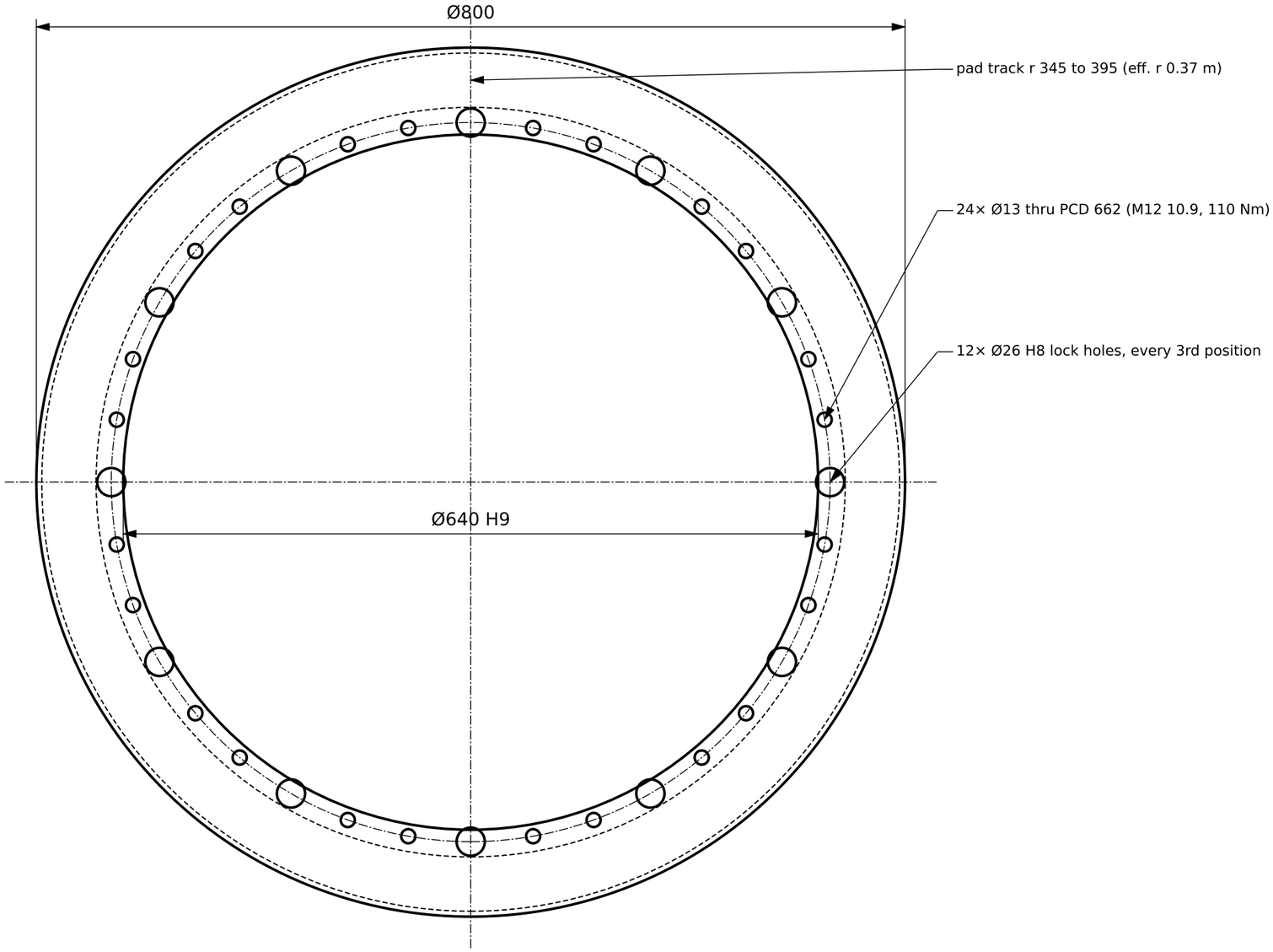
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

MAGNET SEGMENT (PURCHASE DRAWING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-220	A · 2026-11-02	4:1 / 2:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
NdFeB N40UH, NiCuNi + epoxy coating	41 g each	Claude / not checked

VIEW · 36 positions at 10° on PCD 662 · thickness 16



NOTES

1. Faces ground parallel 0.03, flatness 0.05, Ra 1.6. Axial run-out mounted on the drum ≤ 0.10.
2. Two spring-applied hydraulically released calipers (bought), at 90° and 270° (horizontal), each ≥ 15 kN clamp force on 16 mm disc; brake torque 8.7 kNm at μ 0.4.
3. Corrosion: zinc-flake coating outside the pad track; pad track left bare (it is cleaned by the weekly brake test, doc 11).

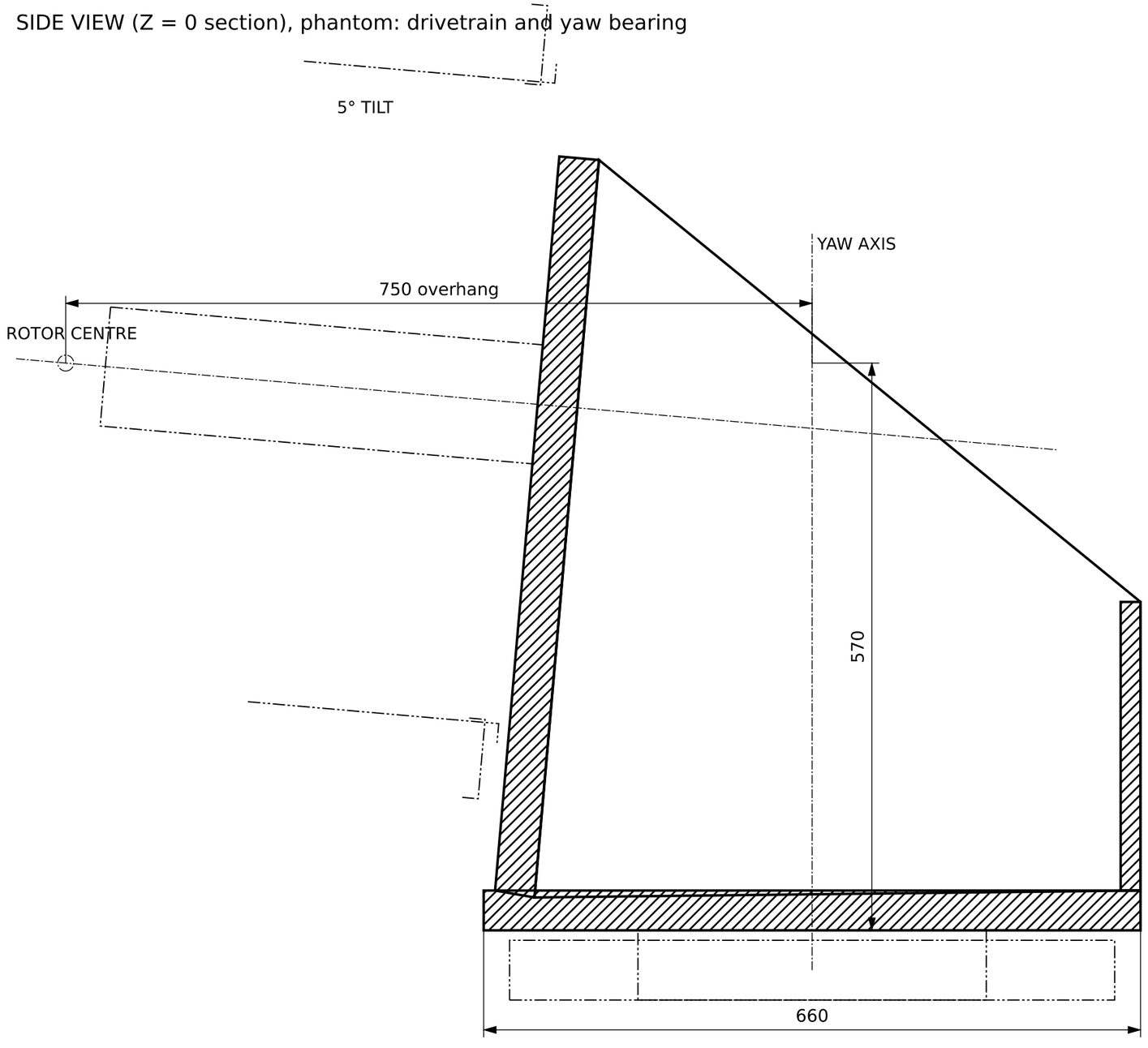
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

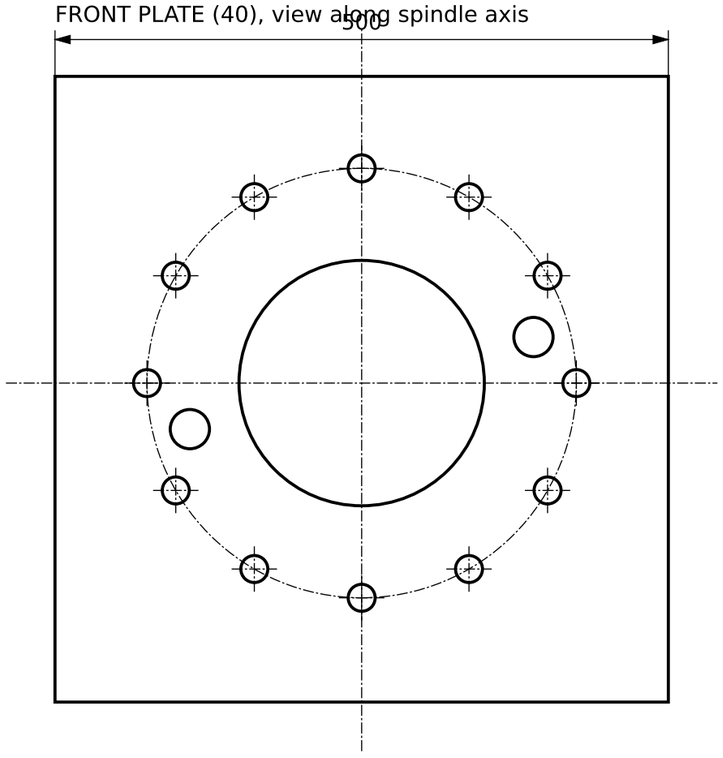
BRAKE DISC AND ROTOR LOCK HOLES

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-230	A · 2026-11-02	1:4.8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C45E+N (or S355J2+N), 16 mm	22.7 kg	Claude / not checked

SIDE VIEW (Z = 0 section), phantom: drivetrain and yaw bearing



FRONT PLATE (40), view along spindle axis



NOTES

1. P1 bedplate is a stress-relieved S355J2+N weldment instead of the EN-GJS-400-18U-LT casting (doc 07): no pattern lead time, same interfaces. The cast version follows after P1 strain-gauge data.
2. Front plate 40, perpendicular to the spindle axis: Ø200 H7 spigot bore, 12× Ø22 PCD 350 (M20 10.9 from inside, 540 Nm at μ 0.12), 2× Ø32 cable holes PCD 290. Face flatness 0.03, square to the yaw flange within 0.05/400 at 5.00° ±0.05° tilt.
3. Bottom plate 40 (yaw flange): Ø220 bore, 16× Ø18 PCD 300 for the yaw bearing inner ring (M16 10.9). Flatness 0.05 (slewing ring requirement, confirm with supplier).
4. Side plates 20 (Z = ±240), rear plate 20, internal gusset under the front plate; top is open for access in P1 (bolted lid).
5. Brake caliper brackets NP1-310 bolt to side plates at the spindle axis height; yaw drive and yaw brake brackets bolt to the bottom plate (positions after yaw supplier drawing).
6. Welds ISO 5817 B, 100 % MT; stress relieve; machine front plate and bottom plate in one set-up on a 5° fixture.

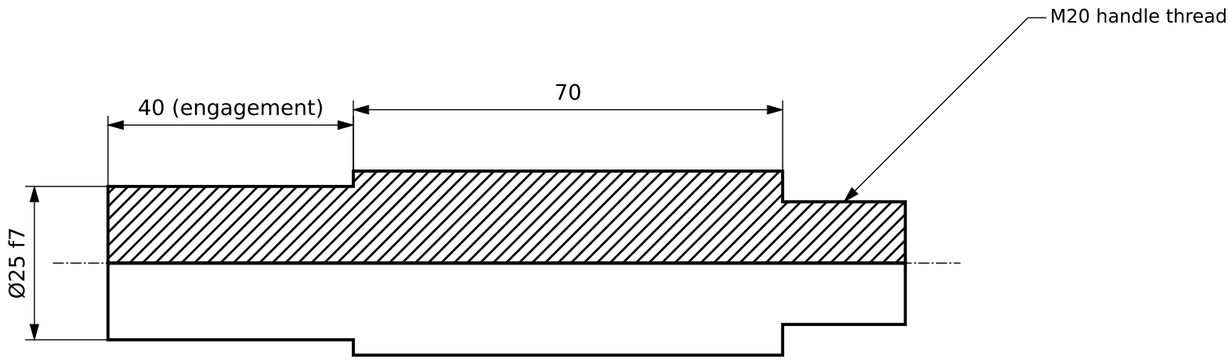
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

BEDPLATE WELDMENT (P1 FABRICATED VERSION)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-300	A · 2026-11-02	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N plates, stress relieved, machined	≈ 150 kg	Claude / not checked

LOCK PIN



NOTES

1. Shear at 8.7 kNm on PCD 662: 26 kN, τ = 54 MPa (allowable 300). Pin goes through the holder bush on the bedplate bracket into a disc lock hole (20 mm) and the ring behind (20 mm).
2. Holder has a proximity switch that reports 'pin engaged' to the safety chain (no rotor release while engaged, doc 11).
3. Insert only at standstill with the brake applied.

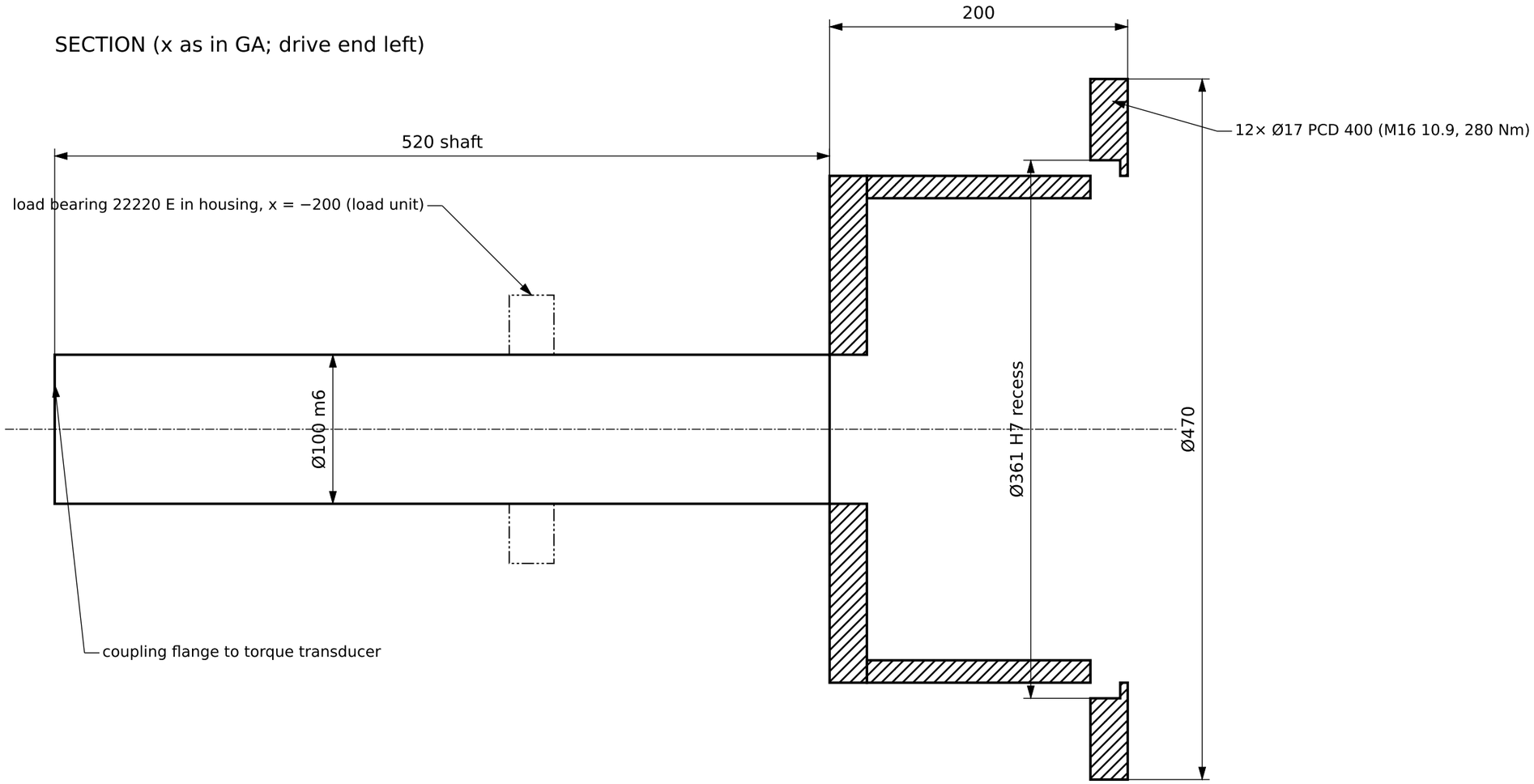
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

ROTOR LOCK PIN AND HOLDER

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-320	A · 2026-11-02	1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
pin 42CrMo4+QT; holder S355J2	≈ 4 kg	Claude / not checked

SECTION (x as in GA; drive end left)



NOTES

1. Replaces the cast hub for P1. Same interface as doc 13: 12× M16 10.9 on PCD 400, Ø360 spigot; it clears the spindle nose, nose cap and front seal cover (min. radial gap 30).
2. Shaft Ø100 forged 42CrMo4+QT, shrink fitted and welded into the front plate; run-out of Ø100 to the recess ≤ 0.05.
3. Radial arm interface (not shown): 4× M24 on the bell at 90° for the 1.0 m torque arm used in brake, lock pin and yaw tests (doc 15).
4. Balanced with the drum to G2.5.

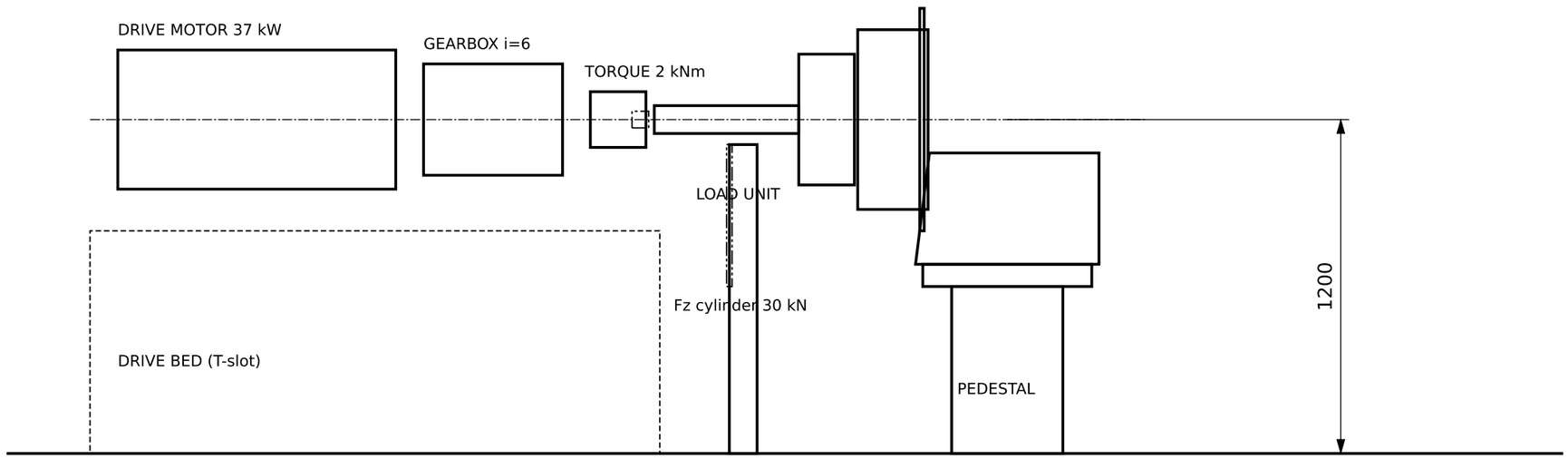
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

BENCH DUMMY HUB (DRIVE AND LOAD ADAPTER)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-600	A · 2026-11-02	1:3.3
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N weldment, stress relieved	≈ 75 kg	Claude / not checked

SIDE VIEW · spindle horizontal (5° wedge under the yaw bearing)



NOTES

1. Energy is recirculated: drive inverter and generator-side load inverter share one DC bus; the grid supplies only the losses (≈ 3 kW at rated).
2. Load unit: spherical roller bearing on the dummy hub shaft at x = −200, pushed by a vertical cylinder (≤ 30 kN, servo-hydraulic) and an axial cylinder (≤ 45 kN, static tests only). Couplings each side of the torque transducer decouple bending from the torque reading.
3. Pedestal, load frame and drive bed bolted to a strong floor or a 6 t concrete block; first pedestal mode > 25 Hz (well above 250 rpm × 3 = 12.5 Hz).
4. Guards: full enclosure of the rotating drum, coupling guards, magnet hazard zone during assembly, interlocked access door wired into the bench safety relay.

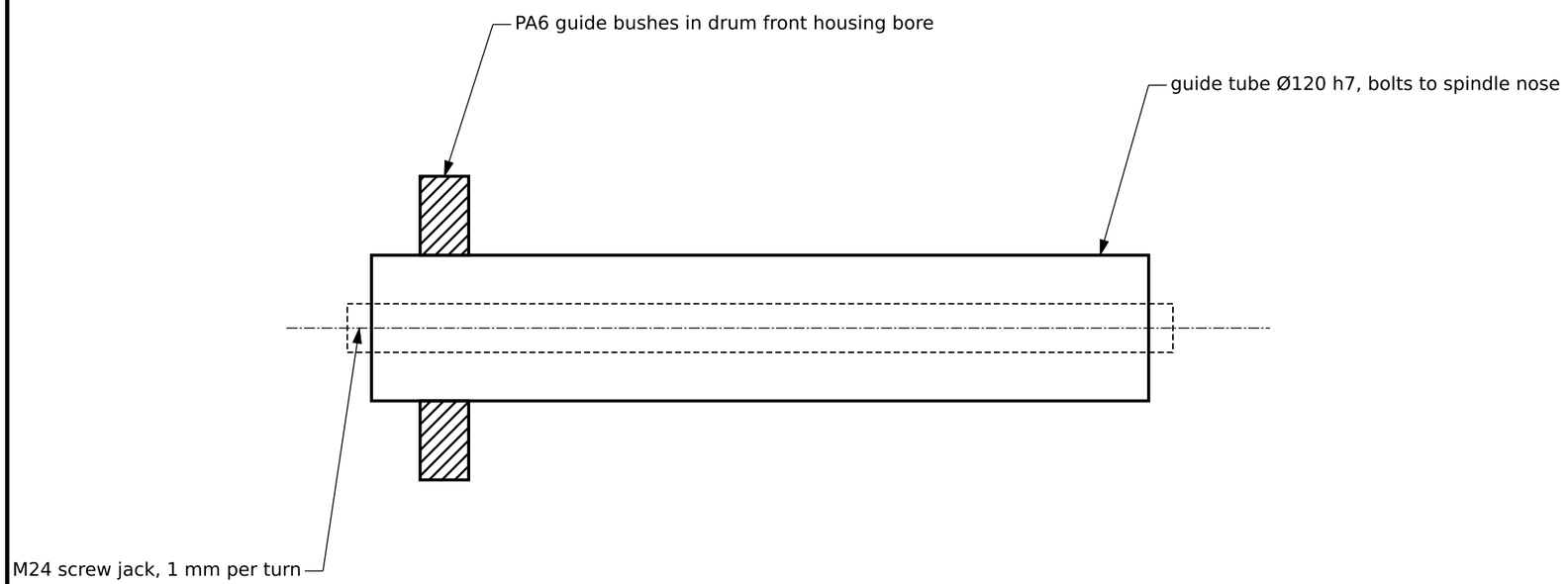
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

BENCH TEST RIG · GENERAL LAYOUT

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-620	A · 2026-11-02	1:20
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
see doc 15		Claude / not checked

GUIDE TUBE ON SPINDLE NOSE



NOTES

1. Radial magnetic pull with the drum off-centre is ≈ 1.9 kN per mm of eccentricity (doc 07) and grows quickly; never let the magnet ring approach the stator by hand.
2. Procedure: spindle vertical, nose up; guide tube bolted to the nose; drum hung from the crane, centred by two PA6 bushes on the tube before the magnets reach the stator; lowered by the screw jack until the rear cup reaches the rear cone.
3. Gap check during lowering: 4 non-magnetic feelers (G10, 1.5 mm) in the gap at 90°.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

DRUM ASSEMBLY GUIDE (MAGNET SAFETY TOOL)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-T01	A · 2026-11-02	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355 / PA6 bushes	≈ 30 kg	Claude / not checked